

7 Growth

Facts in long-run macroeconomic data

1. The growth rate of output γ_y is constant over time.
2. The capital-labor ratio K/L grows at a constant rate.
3. The capital-income ratio K/y is constant.
4. Capital and labor shares of income are constant.
5. Real rates of return are constant.

7.1 Feasibility of growth

The standard case is that of labor augmenting technological change:

$$c_t + i_t = F_t(K_t; n_t) = F(K_t; \gamma^t n_t),$$

where F represents a constant returns to scale production technology and $\gamma > 1$.

Our object of study is what is called balanced growth: all economic variables grow at constant rates (that could vary from one variable to another).

$$y_t = y_0 g_y^t$$

$$c_t = c_0 g_c^t$$

$$K_t = K_0 g_K^t$$

$$i_t = i_0 g_i^t$$

$$n_t = n_0 g_n^t$$

First g_n cannot exceed 1. Hence $g_n = 1$.

From the capital accumulation law

$$K_{t+1} = (1 - \delta)K_t + i_t,$$

If both i_t and K_t are to grow at a constant rate, it must be the case that they both grow at the same rate, i.e. $g_K = g_i$. By the same type of reasoning, from the resource constraint

$$c_t + i_t = y_t,$$

we must have that $g_y = g_c = g_i = g_K$. Next, using the fact that F represents a constant-returns-to-scale technology, we have that

$$F(K_t, n_t \gamma^t) = y_t = y_0 g_y^t = F(K_0, n_0) g_y^t = F(K_t, n_0 g_K^t)$$

Hence we have $\gamma = g_K$. All other variables also grow at rate γ .

7.2 The nature of technological change

From the analysis in the previous section, it seems natural to ask whether the assumption that the technological change is labor augmenting is relevant or not. First, what other kinds of technological change can we think of? Let us write the economy's resource constraint, with all the possible types of technological progress that the literature has talked about:

$$c_t + \frac{i_t}{\gamma_i^t} = \gamma_z^t F(\gamma_K^t K_t, \gamma_n^t n_t)$$

THEOREM 10 *If production function is Cobb-Douglass, all the above technology changes are equivalent to labor-augmenting growth after change of variables.*

7.3 Choosing growth

Hence the problem faced by a central planner will be of the form:

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t, n_t), \\ \text{s.t.} \quad & c_t + k_{t+1} \leq F(k_t, \gamma^t n_t) + (1 - \delta)k_t \\ & n_t \in [0, 1] \\ & k_0 > 0 \text{ is given.} \end{aligned}$$

For this type of preference relations, we have the following result:

THEOREM 11 *Balanced growth is possible as a solution to the central planner's problem if and only if*

$$u(c; n) = c^{1-\sigma} v(1 - n),$$

where time endowment is normalized to one as usual and $v(\cdot)$ is a function with leisure as an argument.

7.4 Transforming the model

We define transformed variables by dividing each original variable by its growth rate:

$$\begin{aligned} \hat{c}_t &= \frac{c_t}{\gamma^t} \\ \hat{i}_t &= \frac{i_t}{\gamma^t} \\ \hat{K}_t &= \frac{K_t}{\gamma^t} \end{aligned}$$

and thus obtain the transformed model ($\hat{\beta} = \beta\gamma^{1-\sigma}$):

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \hat{\beta}^t \hat{c}_t^{1-\sigma} v(1 - n_t), \\ \text{s.t.} \quad & \hat{c}_t + \gamma \hat{k}_{t+1} \leq F(\hat{k}_t, n_t) + (1 - \delta) \hat{k}_t \\ & n_t \in [0, 1] \\ & k_0 > 0 \text{ is given.} \end{aligned}$$

We need to check the conditions for this problem to be well defined. This requires that $\beta\gamma^{1-\sigma} < 1$.

Therefore with the stated assumptions on preferences and on technology, the model converges to a balanced growth path, in which all variables grow at rate γ . This rate is exogenously determined; it is a parameter in the model. That is the reason why it is called *exogenous* growth model.